



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Quadratic Equations

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Your notes

Solving Quadratics by Factorising

Solving Quadratics by Factorising

How do I solve a quadratic equation using factorisation?

- **Rearrange** it into the form $ax^2 + bx + c = 0$
 - Zero must be on one side
 - It is easier if you rearrange so that a is positive
- **Factorise** the quadratic and **solve** each bracket equal to zero
 - If $(x + 4)(x - 1) = 0$, then either $x + 4 = 0$ or $x - 1 = 0$
 - Because if two things multiply together to give zero,
 - then one or the other of them must be equal to zero
- To solve $(x - 3)(x + 7) = 0$
 - ...solve **first bracket = 0**:
 - $x - 3 = 0$
 - add 3 to both sides: $x = 3$
 - ...and solve **second bracket = 0**
 - $x + 7 = 0$
 - subtract 7 from both sides: $x = -7$
 - The **two solutions** are $x = 3$ or $x = -7$
 - The solutions in this example are the numbers in the brackets, but with opposite signs

What if there are numbers in front of the x's in the brackets?

- The process is **the same**
 - There's a bit more work to find the solutions
 - You can't just write down the answers by changing the signs
- To solve $(2x - 3)(3x + 5) = 0$
 - ...solve **first bracket = 0**



Your notes

- $2x - 3 = 0$
- add 3 to both sides: $2x = 3$
- divide both sides by 2: $x = \frac{3}{2}$
- ...solve **second bracket = 0**
- $3x + 5 = 0$
- subtract 5 from both sides: $3x = -5$
- divide both sides by 3: $x = -\frac{5}{3}$
- The **two solutions** are $x = \frac{3}{2}$ or $x = -\frac{5}{3}$

What if x is a factor?

- The process is **the same**
 - Just be sure to handle the x correctly
 - That 'x as a factor' gives one of the solutions
- To solve $x(x - 4) = 0$
 - it may help to think of x as $(x - 0)$ or (x)
 - ...solve **first bracket = 0**
 - $(x) = 0$, so $x = 0$
 - ...solve **second bracket = 0**
 - $x - 4 = 0$
 - add 4 to both sides: $x = 4$
 - The **two solutions** are $x = 0$ or $x = 4$
- It is a common **mistake** to divide (cancel) both sides by x at the beginning
 - If you do this you will **lose a solution** (the $x = 0$ solution)

How can I use my calculator to help with solving quadratics by factorising?



Your notes

- You can use your calculator to **help you to factorise**

- A calculator gives solutions to $6x^2 + x - 2 = 0$ as $x = -\frac{2}{3}$ and $x = \frac{1}{2}$

- **Reverse** the method above to factorise!

- $6x^2 + x - 2 = (3x + 2)(2x - 1)$

- **Be careful:** a calculator also gives solutions to $12x^2 + 2x - 4 = 0$ as $x = -\frac{2}{3}$ and $x = \frac{1}{2}$

- But $12x^2 + 2x - 4 \neq (3x + 2)(2x - 1)$

- The right-hand side expands to $6x^2 + \dots$ not $12x^2 + \dots$

- **Multiply** outside the brackets by 2 to correct this

- $12x^2 + 2x - 4 = 2(3x + 2)(2x - 1)$



Examiner Tips and Tricks

- Remember that you can check your solutions by either
 - substituting them back into the original equation
 - using a different quadratic method
 - or using a calculator



Worked Example

(a) Solve $(x - 2)(x + 5) = 0$

Set the first bracket equal to zero

$$x - 2 = 0$$

Add 2 to both sides

$$x = 2$$

Set the second bracket equal to zero



Your notes

$$x + 5 = 0$$

Subtract 5 from both sides

$$x = -5$$

Write both solutions together using "or"

$$x = 2 \text{ or } x = -5$$

(b) Solve $(8x + 7)(2x - 3) = 0$

Set the first bracket equal to zero

$$8x + 7 = 0$$

Subtract 7 from both sides

$$8x = -7$$

Divide both sides by 8

$$x = -\frac{7}{8}$$

Set the second bracket equal to zero

$$2x - 3 = 0$$

Add 3 to both sides

$$2x = 3$$

Divide both sides by 2

$$x = \frac{3}{2}$$

Write both solutions together using "or"

$$x = -\frac{7}{8} \text{ or } x = \frac{3}{2}$$

(c) Solve $x(5x - 1) = 0$

Do not divide both sides by x (this will lose a solution at the end)

Set the first "bracket" equal to zero

$$(x) = 0$$



Your notes

Solve this equation to find x

$$x = 0$$

Set the second bracket equal to zero

$$5x - 1 = 0$$

Add 1 to both sides

$$5x = 1$$

Divide both sides by 5

$$x = \frac{1}{5}$$

Write both solutions together using “or”

$$x = 0 \text{ or } x = \frac{1}{5}$$



Your notes

The Quadratic Formula

Quadratic Formula

What is the quadratic formula?

- A **quadratic equation** has the form $ax^2 + bx + c = 0$ (where $a \neq 0$)
 - you need " $= 0$ " on one side
- The **quadratic formula** is a formula that gives both solutions to a quadratic equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



Examiner Tips and Tricks

- Make sure the quadratic equation has " $= 0$ " on the right-hand side
 - Otherwise it needs rearranging first

How do I use the quadratic formula to solve a quadratic equation?

- Read off the **values** of a , b and c from the equation
- **Substitute** these into the formula
 - **Write** this line of working in the exam
 - Put **brackets** around any **negative numbers** being substituted in
- To solve $2x^2 - 8x - 3 = 0$ using the quadratic formula:
 - $a = 2$, $b = -8$ and $c = -3$

- $x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4 \times 2 \times (-3)}}{2 \times 2}$

- Either type this into a calculator or simplify by hand
 - Type it once using $+$ for $\pm$ then again using $-$ for $\pm$



Your notes

- The **solutions** are $x = 4.3452078...$ or $x = -0.34520787...$
 - To **3 decimal places**: $x = 4.345$ or $x = -0.345$
 - To **3 significant figures**: $x = 4.35$ or $x = -0.345$



Examiner Tips and Tricks

- Always look for how the question wants you to leave your final answers
 - For example, correct to 2 decimal places

How do I write the solutions in an exact (surd) form?

- You may be asked to give answers in an **exact (surd)** form
- In the example above, work out the **number** under the **square root** sign

- Be careful with negatives!

- $(-8)^2 - 4 \times 2 \times (-3) = 64 + 24 = 88$

- Now square root this number and use **surd rules** to simplify

- $\sqrt{88} = \sqrt{4 \times 22} = \sqrt{4} \times \sqrt{22} = 2\sqrt{22}$

- **Substitute** this back into the formula and simplify

- $$x = \frac{8 \pm 2\sqrt{22}}{4} = \frac{2(4 \pm \sqrt{22})}{4} = \frac{4 \pm \sqrt{22}}{2}$$

- The solutions in **exact (surd)** form are $x = \frac{4 + \sqrt{22}}{2}$ or $x = \frac{4 - \sqrt{22}}{2}$

- **Calculators** that can solve quadratics will give solutions in **exact (surd)** form

What is the discriminant?

- The part of the formula under the square root ($b^2 - 4ac$) is called the **discriminant**
- The **sign** of this value tells you if there are 0, 1 or 2 solutions
 - If $b^2 - 4ac > 0$ (positive)
 - then there are 2 different solutions



Your notes

- If $b^2 - 4ac = 0$
 - then there is only 1 solution
 - sometimes called "two repeated solutions"
- If $b^2 - 4ac < 0$ (negative)
 - then there are no solutions
 - If your calculator gives you solutions with i terms in, these are "complex" and are not what we are looking for
- Interestingly, if $b^2 - 4ac$ is a perfect square number (1, 4, 9, 16, ...) then the quadratic expression could have been factorised!

Can I use my calculator to solve quadratic equations?

- If your calculator solves quadratic equations, use it to **check** your **final answers**
 - But a correct method and working must still be shown



Worked Example

Use the quadratic formula to find the solutions of the equation $3x^2 - 2x - 4 = 0$.
Give each solution as an exact value in its simplest form.

Write down the values of a , b and c

$$a = 3, b = -2, c = -4$$

Substitute these values into the quadratic formula, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Put brackets around any negative numbers

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times 3 \times (-4)}}{2 \times 3}$$

Simplify the expressions

$$x = \frac{2 \pm \sqrt{4 + 48}}{6} = \frac{2 \pm \sqrt{52}}{6}$$

Simplify the surd



Your notes

$$x = \frac{2 \pm \sqrt{4 \times 13}}{6} = \frac{2 \pm 2\sqrt{13}}{6}$$

Simplify the fraction

$$x = \frac{1 \pm \sqrt{13}}{3}$$



Your notes

Completing the Square

Completing the Square

How can I rewrite the first two terms of a quadratic expression as the difference of two squares?

- Look at the quadratic expression $x^2 + bx + c$
- The first **two terms** can be written as the **difference of two squares** using the following rule

$x^2 + bx$ is the same as $(x + p)^2 - p^2$ where p is **half** of b

- Check this is true by expanding the right-hand side
 - Is $x^2 + 2x$ the same as $(x + 1)^2 - 1^2$?
 - Yes: $(x + 1)(x + 1) - 1^2 = x^2 + 2x + 1 - 1 = x^2 + 2x$
- This works for **negative** values of b too
 - $x^2 - 20x$ can be written as $(x - 10)^2 - (-10)^2$ which is $(x - 10)^2 - 100$
 - A negative b does not change the sign at the end

How do I complete the square?

- Completing the square** is a way to rewrite a quadratic expression in a form containing a **squared bracket**
- To complete the square on $x^2 + 10x + 9$
 - Use the rule above to replace the first two terms, $x^2 + 10x$, with $(x + 5)^2 - 5^2$
 - then add 9: $(x + 5)^2 - 5^2 + 9$
 - simplify the numbers:** $(x + 5)^2 - 25 + 9$
 - answer: $(x + 5)^2 - 16$

How do I complete the square when there is a coefficient in front of the x^2 term?

- You first need to take **a** out as a **factor** of the x^2 and x terms only
 - Factorise the first two terms**



Your notes

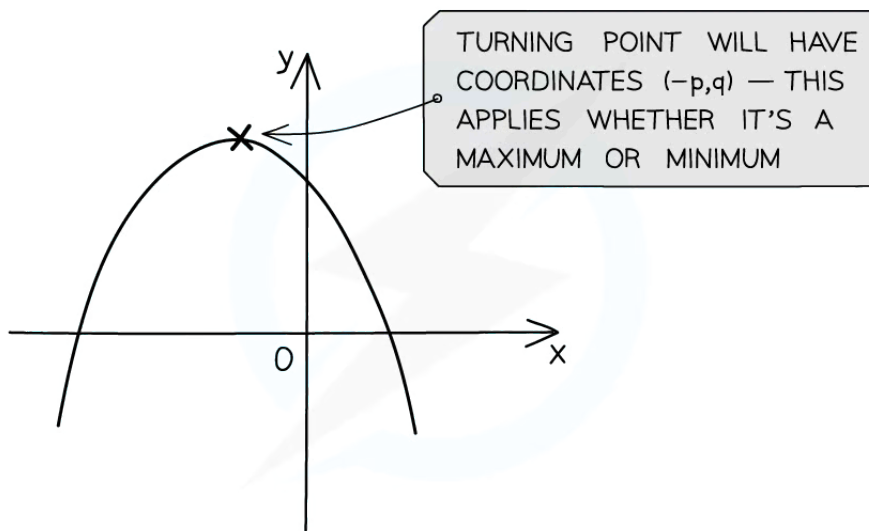
- $ax^2 + bx + c = a\left[x^2 + \frac{b}{a}x\right] + c$
 - Use **square-shaped brackets** here to avoid confusion with round brackets later
- Then complete the square on the bit **inside** the brackets: $x^2 + \frac{b}{a}x$
 - This gives $a[(x + p)^2 - p^2] + c$
 - where p is half of $\frac{b}{a}$
- Finally **multiply** this expression through by a (from outside the square brackets) and **add the c** on to the end
 - $a(x + p)^2 - ap^2 + c$
 - This looks far more complicated than it is in practice!
 - Usually you are asked to give your final answer in the form $a(x + p)^2 + q$
 - For example, $y = 4x^2 + 16x + 5$
 - Factorise out ' a ' on the right-hand side (use square brackets)
 - $y = 4[x^2 + 4x] + 5$
 - Replace $x^2 + 4x$ with $(x + 2)^2 - 2^2$ (because $p = \frac{4}{2} = 2$)
 - $y = 4[(x + 2)^2 - 2^2] + 5$
 - Simplify the terms inside the square brackets
 - $y = 4[(x + 2)^2 - 4] + 5$
 - Multiply everything inside the square brackets by 4
 - $y = 4(x + 2)^2 - 16 + 5$
 - Simplify to get the final answer
 - $y = 4(x + 2)^2 - 11$
 - For quadratics like $-x^2 + bx + c$, do the above but with $a = -1$

How do I find the turning point by completing the square?



Your notes

- Completing the square helps us find the **turning point** on a quadratic graph
 - If $y = (x + p)^2 + q$ then the turning point is at $(-p, q)$
 - Notice the negative sign in the x-coordinate
 - This links to transformations of graphs
 - A translation of $y = x^2$ by p to the left and q up
 - If $y = a(x + p)^2 + q$ then the turning point is still at $(-p, q)$
 - The a does **not change** the coordinates
 - The turning point is a minimum point if $a > 0$
 - or a maximum point if $a < 0$
- This can also help you **create the equation** of a quadratic when given the turning point



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- It can also be used to **prove** or **show** results using the fact that any **squared term**, such as the squared bracket $(x \pm p)^2$, will always be **greater than or equal to 0**
 - You cannot square a number and get a negative value
 - The smallest a squared term can be is 0



Your notes

e.g. $y = x^2 + 6x - 3$

$$y = (x+3)^2 - 12$$

$$\therefore y \geq -12$$

A SQUARED TERM
WILL HAVE A MINIMUM
VALUE OF 0

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Examiner Tips and Tricks

- To know if you have completed the square correctly, expand your answer to check



Worked Example

(a) By completing the square, find the coordinates of the turning point on the graph of $y = x^2 + 6x - 11$.

Find half of +6 (call this p)

$$p = \frac{6}{2} = 3$$

Write $x^2 + 6x$ in the form $(x+p)^2 - p^2$

$$x^2 + 6x \text{ is the same as } (x+3)^2 - 3^2$$

Put this result into the equation of the curve

$$y = (x+3)^2 - 3^2 - 11$$

Simplify the numbers

$$y = (x+3)^2 - 20$$



Your notes

Use the fact that the turning point of $y = (x + p)^2 + q$ is at $(-p, q)$

Here $p = 3$ and $q = -20$

turning point at $(-3, -20)$

(b) Write $-3x^2 + 12x + 24$ in the form $a(x + p)^2 + q$.

Factorise -3 out of the first two terms only

Use square-shaped brackets

$$-3[x^2 - 4x] + 24$$

Complete the square on the $x^2 - 4x$ inside the brackets

Write in the form $(x + p)^2 - p^2$ where p is half of -4

$$-3[(x - 2)^2 - (-2)^2] + 24$$

Simplify the numbers inside the brackets

$(-2)^2$ is 4

$$-3[(x - 2)^2 - 4] + 24$$

Multiply -3 by all the terms inside the square brackets

(You do not multiply -3 by the 24)

$$-3(x - 2)^2 + 12 + 24$$

Simplify the numbers

$$-3(x - 2)^2 + 36$$

This is now in the form $a(x + p)^2 + q$ where $a = -3$, $p = -2$ and $q = 36$

$$-3(x - 2)^2 + 36$$

Solving by Completing the Square

How do I solve a quadratic equation by completing the square?

- To solve $x^2 + bx + c = 0$
 - **replace** the first two terms, $x^2 + bx$, with $(x + p)^2 - p^2$ where p is half of b
 - This is **completing the square**
 - $x^2 + bx + c = 0$ becomes $(x + p)^2 - p^2 + c = 0$



Your notes

- (where p is half of b)
- rearrange this equation to **make x the subject** (using $\pm\sqrt{}$)
- For example, solve $x^2 + 10x + 9 = 0$ by completing the square
 - $x^2 + 10x$ becomes $(x + 5)^2 - 5^2$
 - so $x^2 + 10x + 9 = 0$ becomes $(x + 5)^2 - 5^2 + 9 = 0$
 - make x the subject (using $\pm\sqrt{}$)
 - $(x + 5)^2 - 25 + 9 = 0$
 - $(x + 5)^2 = 16$
 - $x + 5 = \pm\sqrt{16}$
 - $x + 5 = \pm 4$
 - $x = -5 \pm 4$
 - $x = -1$ or $x = -9$
- It also works with numbers that lead to **surds**
 - The answers found will be in **exact (surd)** form



Examiner Tips and Tricks

- When making x the subject to find the solutions, don't expand the squared bracket back out again!
 - Remember to use $\pm\sqrt{}$ to get **two** solutions

How do I solve by completing the square when there is a coefficient in front of the x^2 term?

- If the equation is $ax^2 + bx + c = 0$ with a **number** (other than 1) in front of x^2
 - you can **divide both sides by a** first (before completing the square)
 - For example $3x^2 + 12x + 9 = 0$
 - Divide both sides by 3
 - $x^2 + 4x + 3 = 0$
 - Complete the square on this easier equation



Your notes

- This trick **only** works when completing the square to **solve** a quadratic equation
 - i.e. it has an "=0" on the right-hand side
- **Don't** do this when using completing the square to rewrite a **quadratic expression** in a new form
 - i.e. when there is no "=0"
 - For that, you must **factorise out** the a (but not divide by it)

$$ax^2 + bx + c = a \left[x^2 + \frac{b}{a}x \right] + c \text{ and so on}$$

How does completing the square link to the quadratic formula?

- The **quadratic formula** actually comes from **completing the square** to solve $ax^2 + bx + c = 0$
 - a, b and c are left as letters when completing the square
 - This makes it as general as possible
- You can see hints of this when you solve quadratics
 - For example, solving $x^2 + 10x + 9 = 0$
 - by completing the square, $(x + 5)^2 = 16$ so $x = -5 \pm 4$ (as above)
 - by the quadratic formula, $x = \frac{-10 \pm \sqrt{64}}{2} = -5 \pm \frac{8}{2} = -5 \pm 4$ (the same structure)



Worked Example

Solve $2x^2 - 8x - 24 = 0$ by completing the square.

Divide both sides by 2 to make the quadratic start with x^2

$$x^2 - 4x - 12 = 0$$

Halve the middle number, -4 , to get -2

Replace the first two terms, $x^2 - 4x$, with $(x - 2)^2 - (-2)^2$

$$(x - 2)^2 - (-2)^2 - 12 = 0$$

Simplify the numbers



Your notes

$$(x-2)^2 - 4 - 12 = 0$$

$$(x-2)^2 - 16 = 0$$

Add 16 to both sides

$$(x-2)^2 = 16$$

Take the square root of both sides

Include the $\pm$ sign to get two solutions

$$x-2 = \pm\sqrt{16} = \pm 4$$

Add 2 to both sides

$$x = 2 \pm 4$$

Work out each solution separately

$$x = 6 \text{ or } x = -2$$



Your notes

Deciding the Quadratic Method

Deciding the Quadratic Method

If you have to solve a quadratic equation but are not told which method to use, here is a guide for what to do.

When should I solve by factorisation?

- Use factorisation when the question asks to **solve by factorisation**
 - For example
 - part (a) Factorise $6x^2 + 7x - 3$
 - part (b) Solve $6x^2 + 7x - 3 = 0$
- Use factorisation when solving **two-term quadratic equations**
 - For example, solve $x^2 - 4x = 0$
 - Take out a **common factor** of x to get $x(x - 4) = 0$
 - So $x = 0$ and $x = 4$
 - For example, solve $x^2 - 9 = 0$
 - Use the **difference of two squares** to factorise it as $(x + 3)(x - 3) = 0$
 - So $x = -3$ and $x = 3$
 - (Or rearrange to $x^2 = 9$ and use $\pm\sqrt{}$ to get $x = \pm 3$)
- Factorising can often be the **quickest** way to solve a quadratic equation

When should I use the quadratic formula?

- Use the quadratic formula when the question says to leave solutions correct to a **given accuracy** (2 decimal places, 3 significant figures etc)
 - This is a hint that the equation will not factorise
- Use the quadratic formula when it may be **faster than factorising**
 - It's quicker to solve $36x^2 + 33x - 20 = 0$ using the quadratic formula than by factorisation
- Use the quadratic formula **if in doubt**, as it **always works**

When should I solve by completing the square?



Your notes

- Use completing the square when part (a) of a question says to **complete the square** and part (b) says to use part (a) to solve the equation
- Use completing the square when making x the **subject of harder formulae** containing both x^2 and x terms
 - For example, make x the subject of the formula $x^2 + 6x = y$
 - Complete the square: $(x + 3)^2 - 9 = y$
 - Add 9 to both sides: $(x + 3)^2 = y + 9$
 - Take square roots and use $\pm$: $x + 3 = \pm \sqrt{y + 9}$
 - Subtract 3: $x = -3 \pm \sqrt{y + 9}$
- Completing the square **always works**
 - But it's not always quick or easy to do



Examiner Tips and Tricks

- If your calculator solves quadratic equations, use it to check your solutions
- If the solutions on your calculator are whole numbers or fractions (with no square roots), this means the quadratic equation does factorise



Worked Example

(a) Solve $x^2 - 7x + 2 = 0$, giving your answers correct to 2 decimal places.

"Correct to 2 decimal places" suggests using the quadratic formula

Substitute $a = 1$, $b = -7$ and $c = 2$ into the formula

Put brackets around any negative numbers

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4 \times 1 \times 2}}{2 \times 1}$$

Use a calculator to find each solution

$$x = 6.70156... \text{ or } 0.2984...$$



Your notes

Round your final answers to 2 decimal places

$$x = 6.70 \text{ or } x = 0.30 \text{ (2 d.p.)}$$

(b) Solve $16x^2 - 82x + 45 = 0$.Method 1

If you cannot spot the factorisation, use the quadratic formula

Substitute $a = 16$, $b = -82$ and $c = 45$ into the formula

Put brackets around any negative numbers

$$x = \frac{-(-82) \pm \sqrt{(-82)^2 - 4 \times 16 \times 45}}{2 \times 16}$$

Use a calculator to find each solution

$$x = \frac{9}{2} \text{ or } x = \frac{5}{8}$$

Method 2If you do spot the factorisation, $(2x - 9)(8x - 5)$, then use that method instead

$$(2x - 9)(8x - 5) = 0$$

Set the first bracket equal to zero

$$2x - 9 = 0$$

Add 9 to both sides then divide by 2

$$2x = 9$$

$$x = \frac{9}{2}$$

Set the second bracket equal to zero

$$8x - 5 = 0$$

Add 5 to both sides then divide by 8

$$8x = 5$$

$$x = \frac{5}{8}$$

$$x = \frac{9}{2} \text{ or } x = \frac{5}{8}$$



Your notes

(c) By writing $x^2 + 6x + 5$ in the form $(x + p)^2 + q$, solve $x^2 + 6x + 5 = 0$.

This question wants you to complete the square first

Find p (by halving the middle number)

$$p = \frac{6}{2} = 3$$

Write $x^2 + 6x$ as $(x + p)^2 - p^2$

$$\begin{aligned} x^2 + 6x &= (x + 3)^2 - 3^2 \\ &= (x + 3)^2 - 9 \end{aligned}$$

Replace $x^2 + 6x$ with $(x + 3)^2 - 9$ in the equation

$$\begin{aligned} (x + 3)^2 - 9 + 5 &= 0 \\ (x + 3)^2 - 4 &= 0 \end{aligned}$$

Now solve it

Make x the subject of the equation (start by adding 4 to both sides)

$$(x + 3)^2 = 4$$

Take square roots of both sides (include a $\pm$ sign to get both solutions)

$$x + 3 = \pm \sqrt{4} = \pm 2$$

Subtract 3 from both sides

$$x = -3 \pm 2$$

Find each solution separately using + first, then - second

$$x = -1 \text{ or } x = -5$$

Even though the quadratic factorises to $(x + 5)(x + 1)$, this is not the method asked for in the question